

SUSHANT LOKHANDE

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EXPERIENCE

San Jose State University | Graduate Research Assistant, Distributed Systems & Data Aug 2025 to May 2026

- Built an automated **telemetry pipeline** that collected, normalized, and analyzed experiment data end to end, cutting setup and analysis effort **55%**.
- Reduced rebuffering **19%** versus a fixed-rate baseline by building a **Docker**, **WebTransport**, and **Media-over-QUIC** testbed and running controlled adaptation experiments across network and protocol configs.
- Validated **14 interoperability scenarios** across networking, dependency, and **failure-recovery** paths to confirm recovery behavior before large experiment runs.
- Generated **136,000** training images from **17,000** binaries across **8** visual representations and benchmarked **6** model families on **1,700** held-out samples.

Turito | Software Engineering Intern, High-Throughput Backend Jun 2022 to Oct 2022

- Lowered network egress **14%** and p95 CPU **22%** on a high-frequency event path by removing redundant processing, payload copies, and per-packet logging overhead.
- Cut controlled-disruption failures **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Reduced production diagnosis time **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.
- Prototyped a **Rust** + **QUIC** media path serving about **8,000** lightweight subscribers with about **24%** lower startup time.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Raised equivalent-load signaling throughput about **31%** by surfacing concurrency bottlenecks that drove targeted backend fixes.
- Cut investigation time about **40%** by building a pipeline that aggregated load-test output from Rust workers into **SQLite** with Python and served comparisons in Flask.

TECHNICAL SKILLS

Languages: Python, Java, C++17, Rust, SQL

Ingestion & Data: Kafka, Hadoop, MapReduce, Hive, PostgreSQL, Redis, SQLite, MongoDB

Distributed Systems: Temporal, gRPC, Tokio, asyncio, WebSockets, Docker, AWS, Linux

Quality: pytest, load testing, profiling, structured metrics, CI/CD

PROJECTS

Melody Metrics, Big-Data Pipeline | C++, Hadoop, MapReduce, Hive, Python, XGBoost

- Processed **1M+** heterogeneous records through C++ **MapReduce** and **Hive** into a curated **140,000**-row dataset powering an **84%**-accurate 81-genre classifier.

AutoGrade AI, Multi-Agent Evaluation Platform | Python, Temporal, Kafka, gRPC, PostgreSQL, Qdrant

- Orchestrated **4-dimension** rubric grading (correctness, efficiency, resilience, quality) across multiple LLM agents on **Temporal** and **Kafka**, with confidence gates and human-review queues.

Relay, LLM Inference Gateway | Python, FastAPI, asyncio, SQLite, SSE

- Cut p50 latency from **759 ms** to **44 ms** over a **4,000**-request load test with an OpenAI-compatible gateway doing exact and **semantic caching**, failover, and rate limiting.

OPEN SOURCE

- **fission/fission**: 3 merged PRs, Kubernetes event and KEDA object capture in support dumps, removed duplicate package enumeration
- **NVIDIA/garak**: preserved attempt UUIDs in report aggregation for traceable LLM evals (merged, shipped in v0.17.0)

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Database Systems, Distributed Computing, NoSQL, Advanced Algorithms, Computer Networks, Machine Learning

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)